Lecture 4: Public Key Encryption and RSA

ITDS271: Computer and Communication Security

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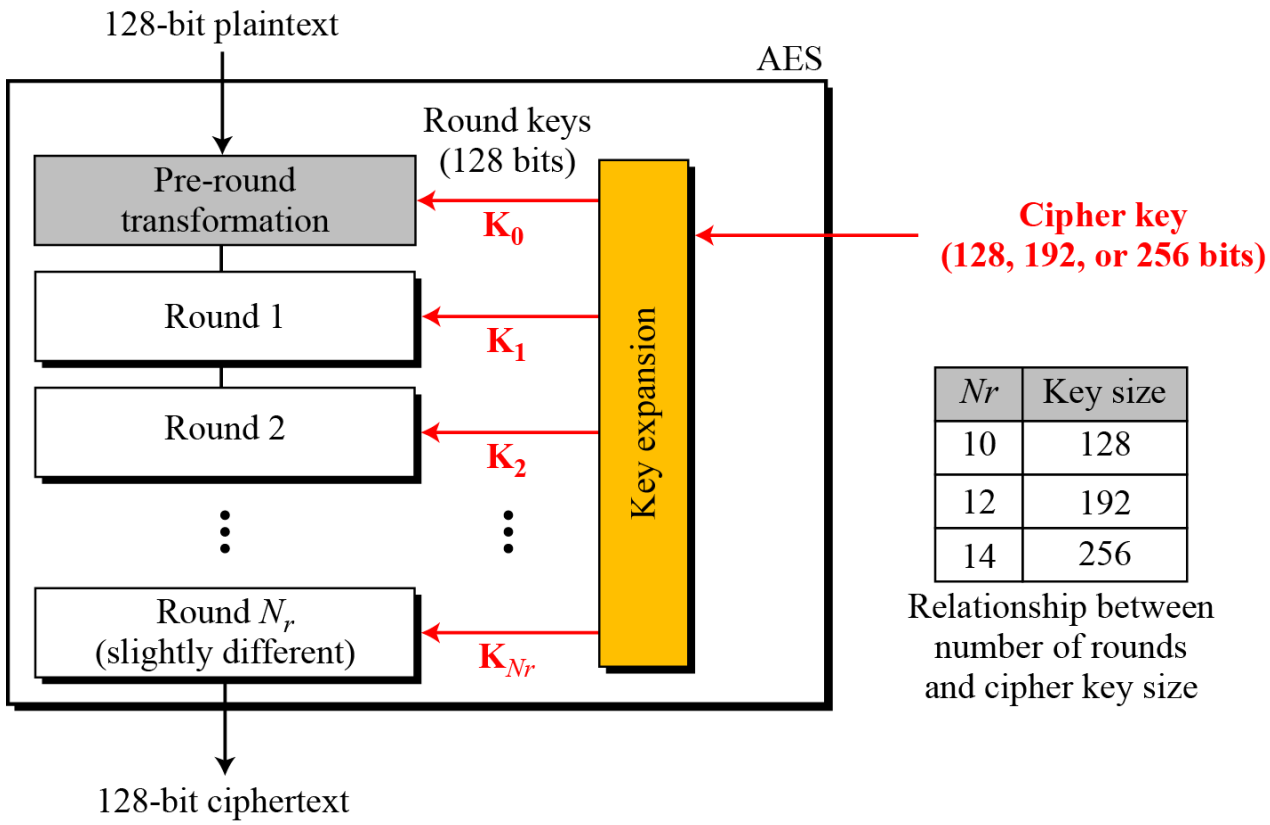
Review: AES

- Symmetric block cipher (**128-bit block**)
- Key size: **128/192/256 bits**
- Based on ***Substitution-Permutation Network (SPN)***
- Iterative process → **round**
  - **Round 1- (N-1):** SubBytes, ShiftRows, MixColumns, AddRoundKey
  - **Round N:** SubBytes, ShiftRows, AddRoundKey



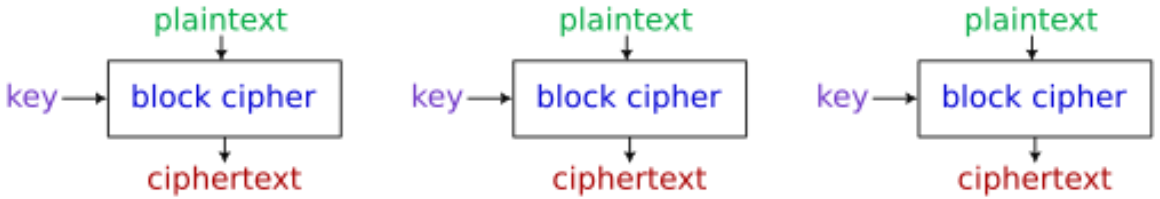
Nr	Key size
10	128
12	192
14	256

Relationship between number of rounds and cipher key size

Review: Mode of Operations

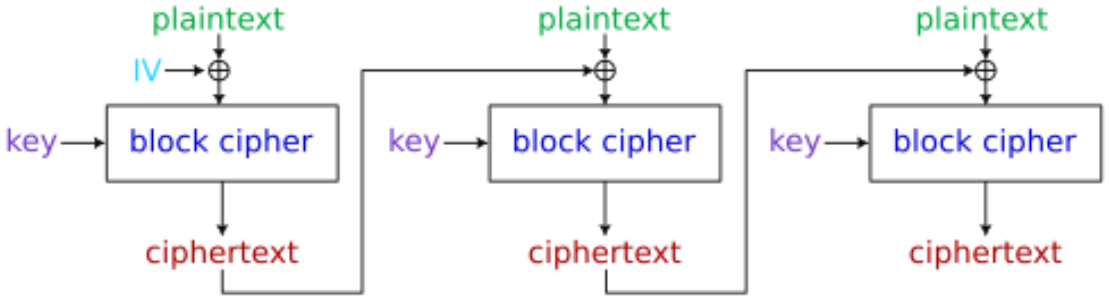
- Handle data that is longer than the block size

Electronic codebook (ECB)



- Simple to implement.
- Allows parallel encryption of blocks (fast)
- Identical plaintext blocks → identical ciphertext blocks.
- Leak patterns

Cipher block chaining (CBC)

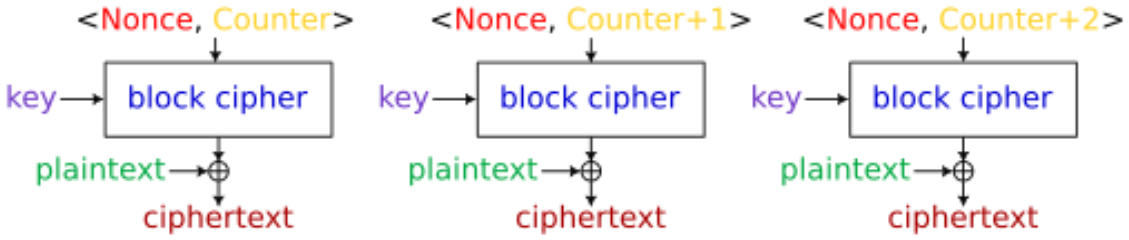


- Chaining removes patterns → more secure than ECB
- Require sequential encryption
- Needs a random IV (must be unique and unpredictable).
- A single-bit error in ciphertext affects two blocks on decryption.

Review: Mode of Operations

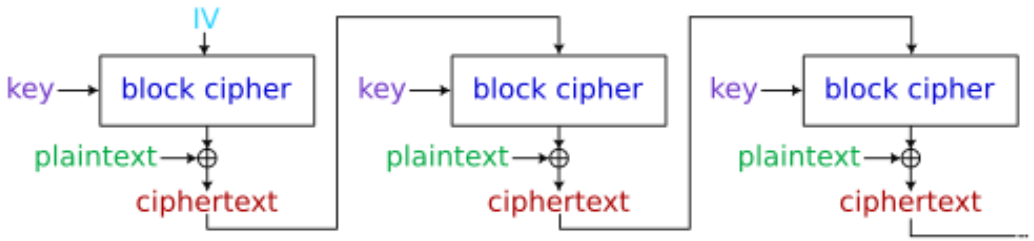
- Stream mode of opertaions
  - Use block ciphers to generate keystream
  - Ciphertext = plaintext XOR keystream

Counter (CTR)



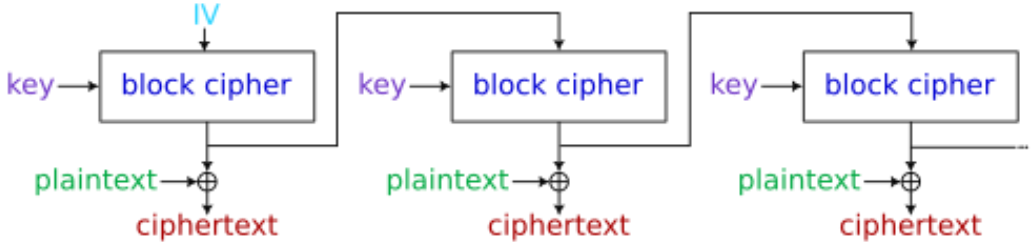
- Fully parallelizable (both encryption & decryption).
- Precomputation of keystream is possible.
- No error propagation
- High efficiency

Cipher feedback (CFB)



- Error propagation
- Sequential encryption

Output feedback (OFB)



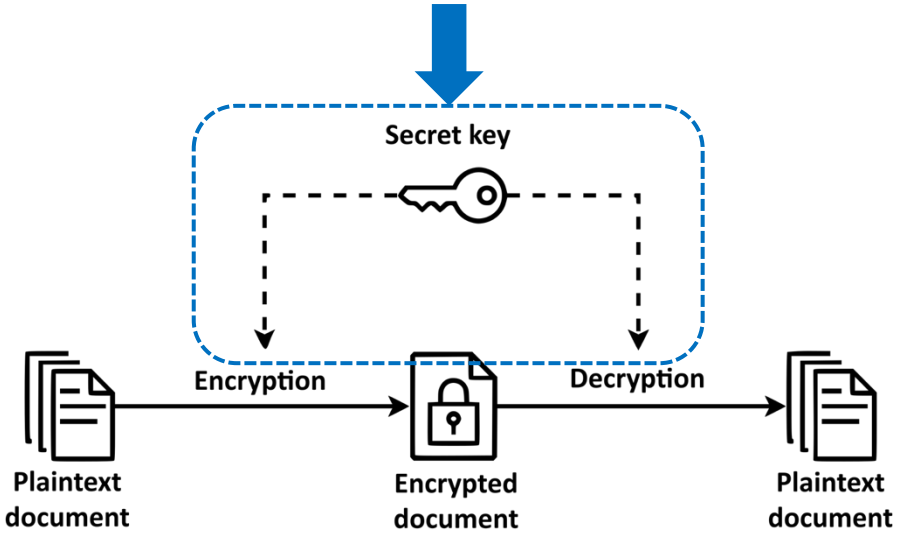
- No error propagation
- Precomputation of keystream is possible.

Topics

- **Public-Key Encryption (PKE)**
- Fundamental Mathematic
- RSA Algorithm
- Key Exchange Algorithm

Drawbacks of Symmetric Key Encryption

- Traditionally, symmetric-key encryption uses <u>one</u> key **shared** by both sender and receiver
- If the key is disclosed, **<u>communications using that key are all compromised</u>**
  - No way to protect sender from the receiver forging a message and claiming that it was sent by the sender.
- ***Key distribution*** (key sharing) is also a problem!



Asymmetric Key Encryption

- Also called **Public-key Encryption (PKE)**
- The most significant advancement in the 3000-year history of cryptography!
- Use clever application of number theoretic concepts to create an encryption function that is easy to calculate but much harder to reverse without sufficient information
- Use as complements rather than replacing symmetric-key encryption

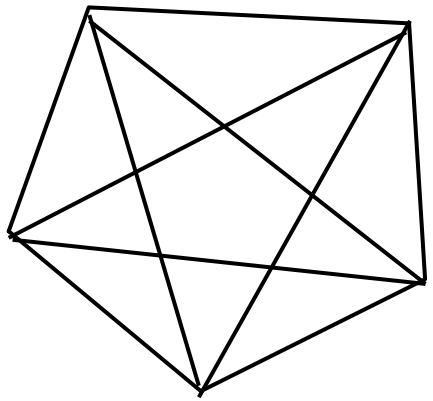
Why PKE?

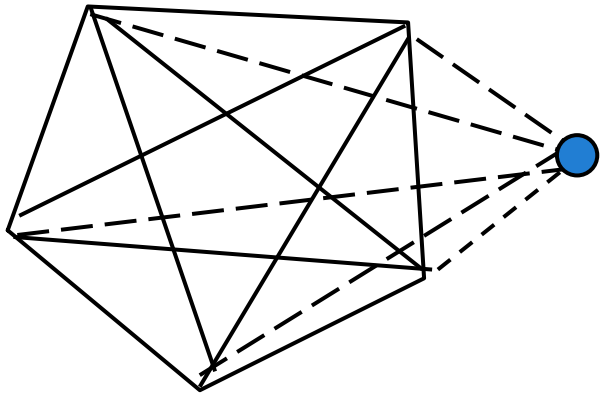
Developed to address 2 main issues:

- **Key distribution:**
  - How to *securely deliver the key* without having to trust a key distribution center (KDC) with your key.
- **Authentication + Non-Repudiation:**
  - How to verify that a message comes intact (without modification) *from the claimed sender*

Key Maintenance Problem

- For conventional system, there must be a common key between any pair of users.
- Therefore, **a lot of keys** must be generated and maintained.
- **Symmetric: *n(n-1)/2* keys** for *n* users
- **Asymmetric (PKE)**: each user needs to maintain only 2 keys.
  - ***Public key*** and ***private key***
  - ***2n* keys** are enough for n users.

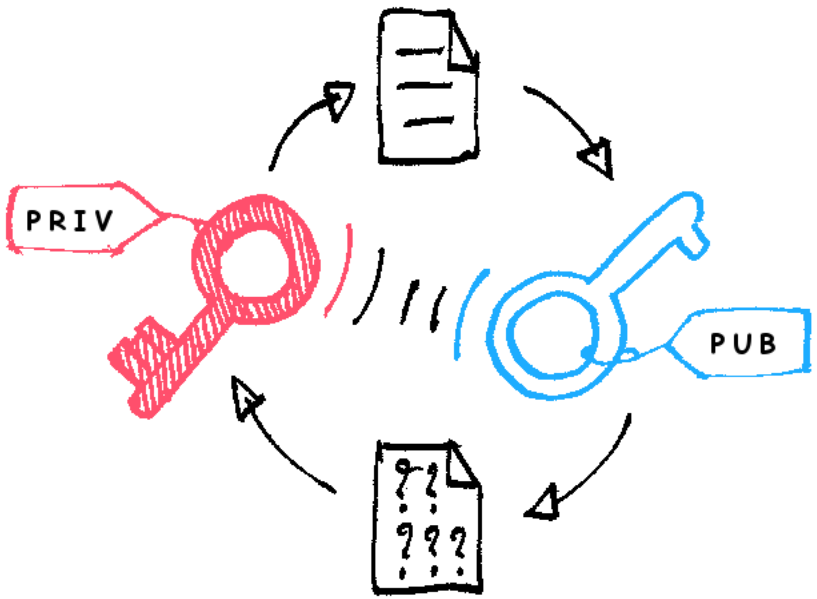




Public Key Encryption

PKE involves 2 keys:

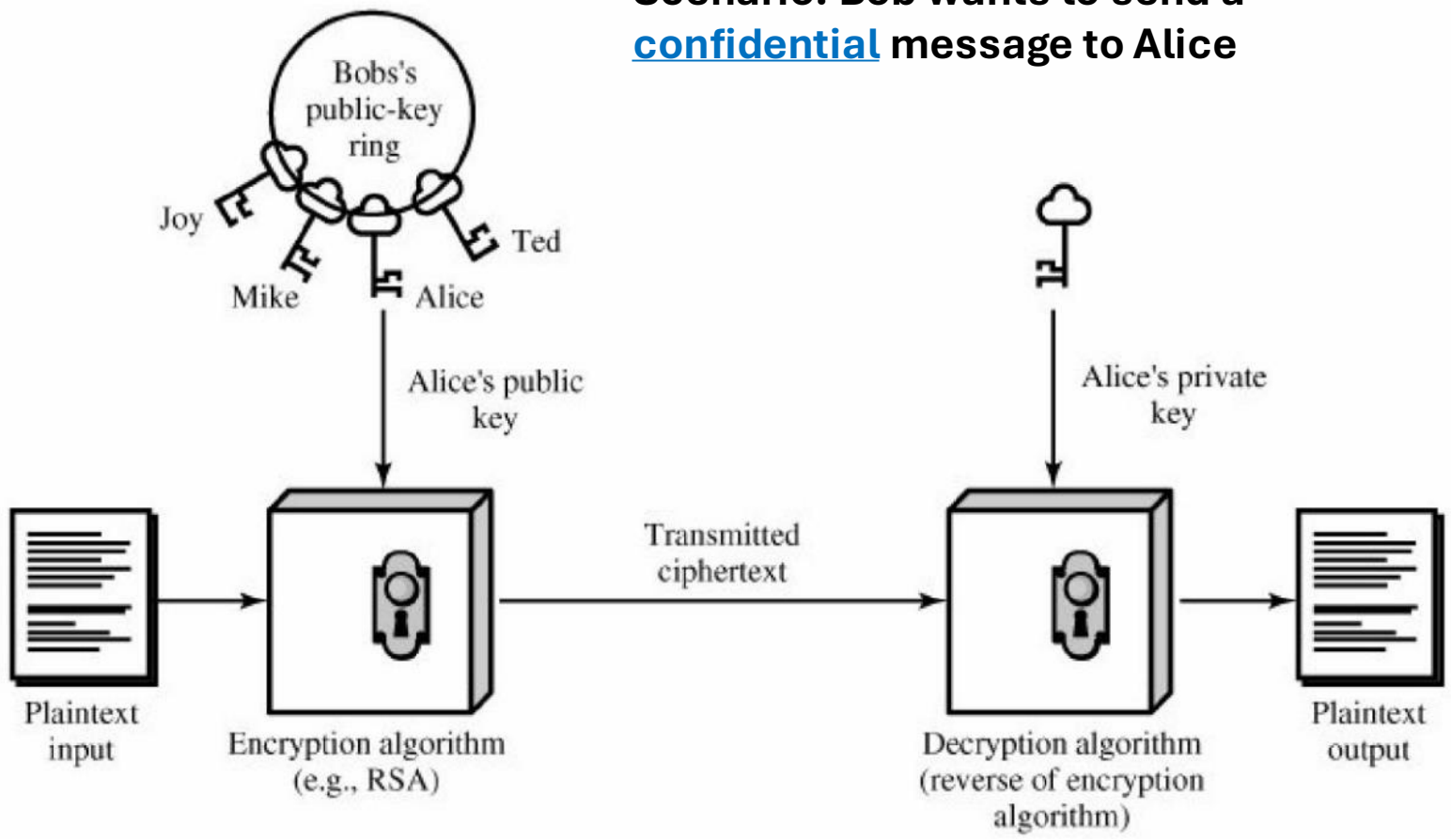
- ***Public-key:***
  - Known by anyone (public)
  - Used for encrypt messages and verify signatures
- ***Private-key:***
  - Known only to the recipient
  - Used to decrypt the message and sign (create) signatures



- It is *asymmetric* because:
  - Those who encrypt the messages or verify the signatures cannot decrypt messages or create signatures.

Public Key Encryption: Data Encryption

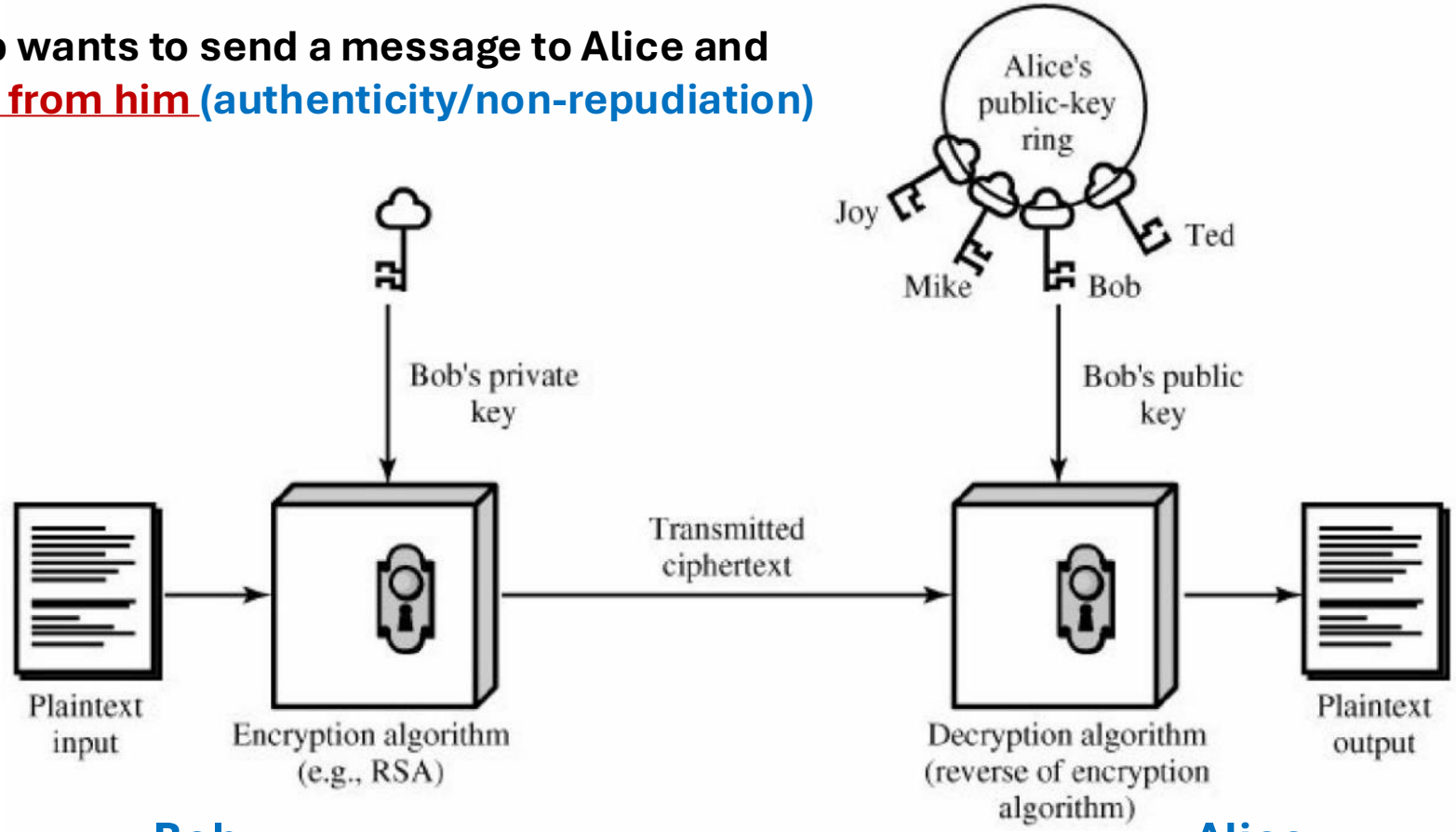
Scenario: Bob wants to send a confidential message to Alice



Bob **Alice**

Public Key Encryption: Digital Signature

Scenario: Bob wants to send a message to Alice and prove it came from him (authenticity/non-repudiation)



Bob **Alice**

Summary of PKE Characteristics

- Solve the **key distribution problem**; no need to transmit a secret key over the network while the public key can be freely distributed
- Highly ***scalable***; suitable for large-scale systems
- Support ***digital signature***
- Based on hard ***mathematical problem***
  - Computationally infeasible to derive the private key from the public key
- Computationally expensive; slower than symmetric key encryption
- Like symmetric-key encryption schemes, brute force (exhaustive) search is always theoretically possible

Topics

- Public-Key Encryption (PKE)
- **Fundamental Mathematic**
- RSA Algorithm
- Key Exchange Algorithm

Prime Number

- Prime number = natural number greater than **1** that
  - Has **exactly two distinct positive divisors; 1 and itself**
  - **1 is not prime** because it has only 1 divisor
- Cannot be written as a product of other numbers
- Examples of Prime Numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, …

Modular Arithmetic

- For any integers, a and n:
  - ***a* mod *n* = *r*** ***r* = remainder when *a* is divided by *n***
  - ***a* = *q**n* + *r*** where $0 \leq r < n$
- All integers can be written: $a = q*n + r$ where $0 < r < n$
  - e.g., 100 mod 11 = 1 → 100 = 9*11 + 1
- Two integers ***a*** and ***b*** are said to be ***congruent modulo n***, if they produce the same remainder when divided by *n*, written as ***a ≡ b mod n***
  - 17 ≡ 5 mod 12
  - -12 ≡ -5 ≡ 2 ≡ 9 ≡ … mod 7

Multiplicative Inverse in Moduluar Arithmetic

- ***x*** is a multiplicative inverse of ***a mod n*** if ***(x*a) ≡ 1 mod n***
- Multiplicative inverse of ***a mod n*** exists if and only if ***a*** and ***n*** are ***coprime***
  - No common divisors apart from 1
  - ***gcd(a,n) = 1***
- For Example:
  - **x = 37** is the inverse of **13 mod 60**
  - Because 13*37 mod 60 = 1

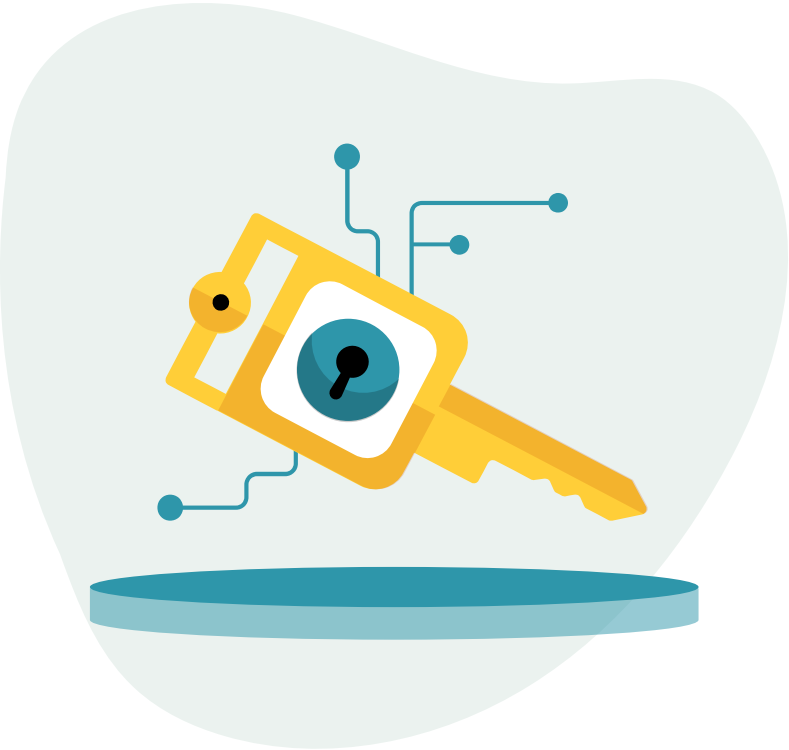
Is 7 a multiplicative inverse of 9 mod 31 ?

Topics

- Public-Key Encryption (PKE)
- Fundamental Mathematic
- **RSA Algorithm**
- Key Exchange Algorithm

RSA

- Invented by **R**ivest, **S**hamir and **A**dleman of MIT in 1978
- Best known and widely used **public-key** encryption scheme
- Security relies on the cost of factoring large prime numbers
  - Factorization problem takes $O(e^{(\log n)(\log \log n)})$ operations → **hard**
- Used for TLS/SSL key exchange, digital signature, code signing



RSA Key Generation

Each user generates a pair of public ***PU = {e,n}*** and private ***PR = {d,n}*** *keys*

1. Randomly choose two large prime number: p and q
2. Compute n (modulus) $n = p \times q$
3. Calculate Euler's Totient function $\phi(n)=(p-1)\times(q-1)$
4. Randomly select an integer **e** (public) satisfying the following conditions:
   - $1 < e < \phi(n)$ and $\gcd(e, \phi(n))=1$
5. Compute **d** (private) where $de \equiv 1 (\mathrm{mod}\, \phi(n))$
   - **d** is the modular multiplicative inverse of **e modulo ϕ(n)**
   - Can be computed using the Extended Euclidean algorithm

Publish the public-key:
PU = {e, n}

Keep the private-key:
PR = {d, n}

RSA Encryption and Decryption

- To <u>encrypt</u> a **message M**, the sender:
  - Obtains the ***public-key of <u>recipient</u>*** — $PU = \{e, n\}$
  - Computes: — $C = M^e \bmod n$, where $0 \leq M < n$
- To <u>decrypt</u> the **ciphertext C**, the recipient:
  - Uses their ***private-key*** — $PR = \{d, n\}$
  - Computes: — $M = C^d \bmod n$
- Note that the message *M* must be smaller than the modulus *n* (separate *M* into blocks if needed)

Example: RSA Key Generation

1)	Select primes:	$p = 17$, $q = 11$
2)	Compute	$n = p * q = 17 * 11 = 187$
3)	Compute	$\emptyset(n) = (p-1)(q-1) = 16 * 10 = 160$
4)	Select e:	$\boldsymbol{gcd(e, 160) = 1}$, **choose** $\boldsymbol{e = 7}$
5)	Determine d:	$\boldsymbol{de \equiv 1 (mod\ \phi(n)),\ d < 160,\ d = 23}$ (since $23 * 7 = 161 = 1 \bmod 160$)
6)	Publish public-key	$\boldsymbol{PU = \{7, 187\}}$
7)	Keep the private-key	$\boldsymbol{PR = \{23, 187\}}$

Example: RSA Encryption and Decryption

Given a message $M = 88$

- **Encryption (PU = {7,187}):**
  - $C = 88^7 \bmod 187 \quad = 11$
- **Decryption (PR = {23, 187}):**
  - $M' = 11^{23} \bmod 187 = 88$

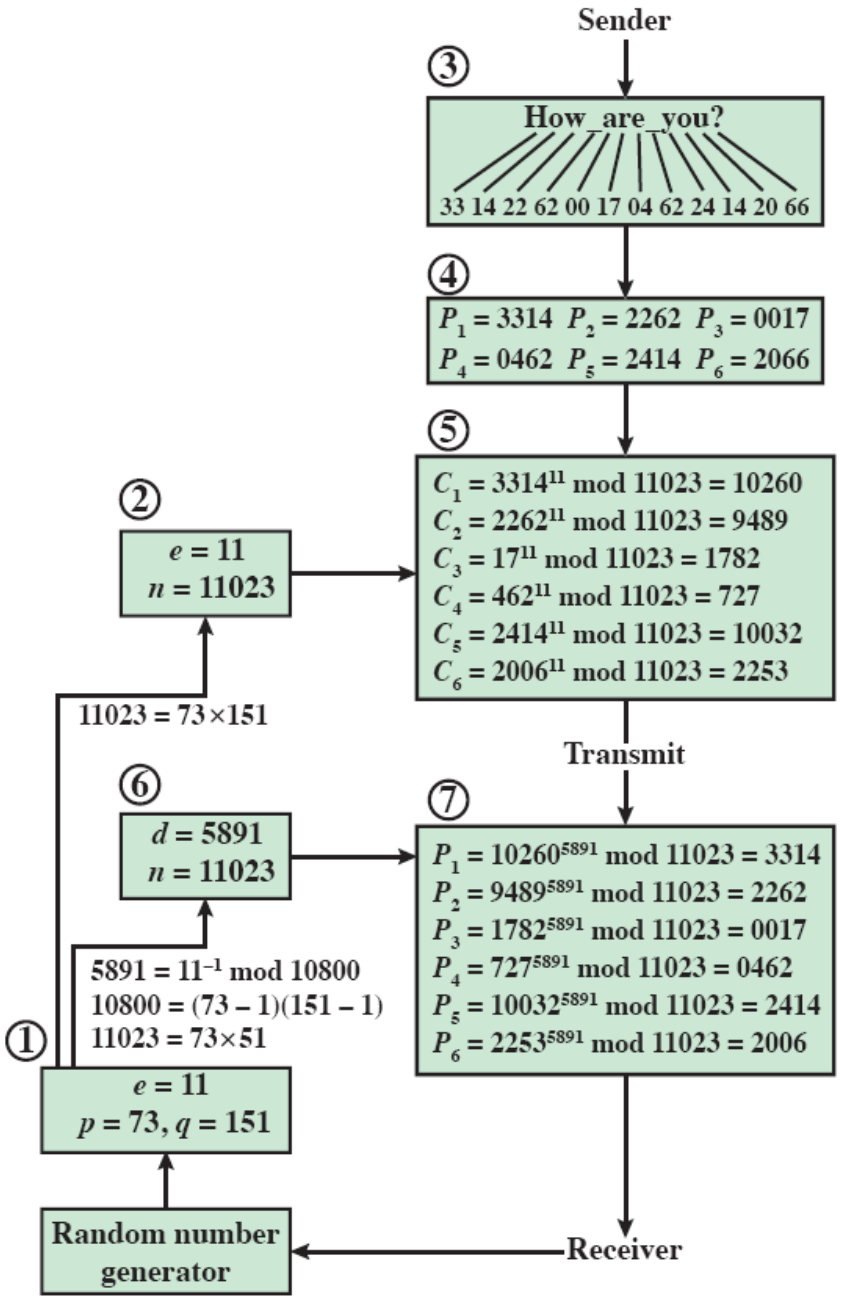
Example: RSA Cryptosystem

$PU = \{e, n\}$

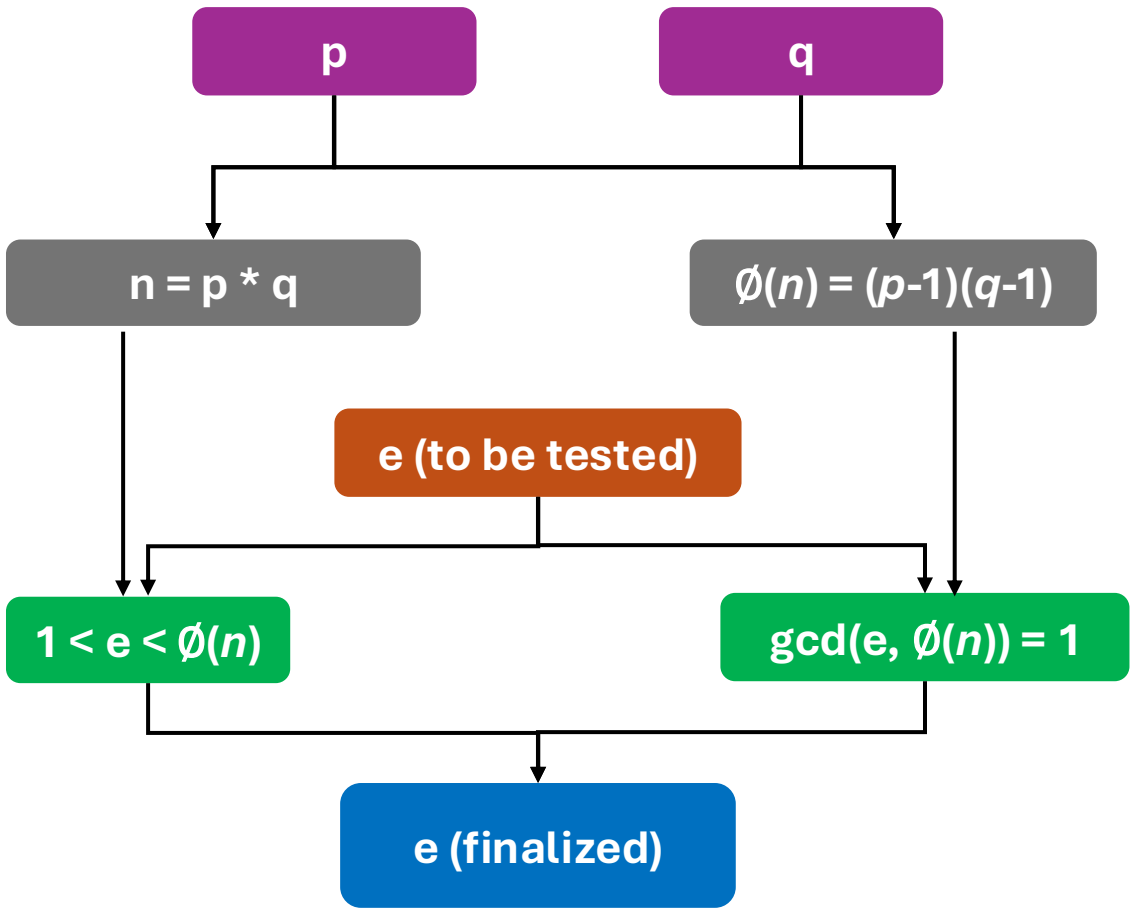
$C = M^e \bmod n$, where $0 \le M < n$

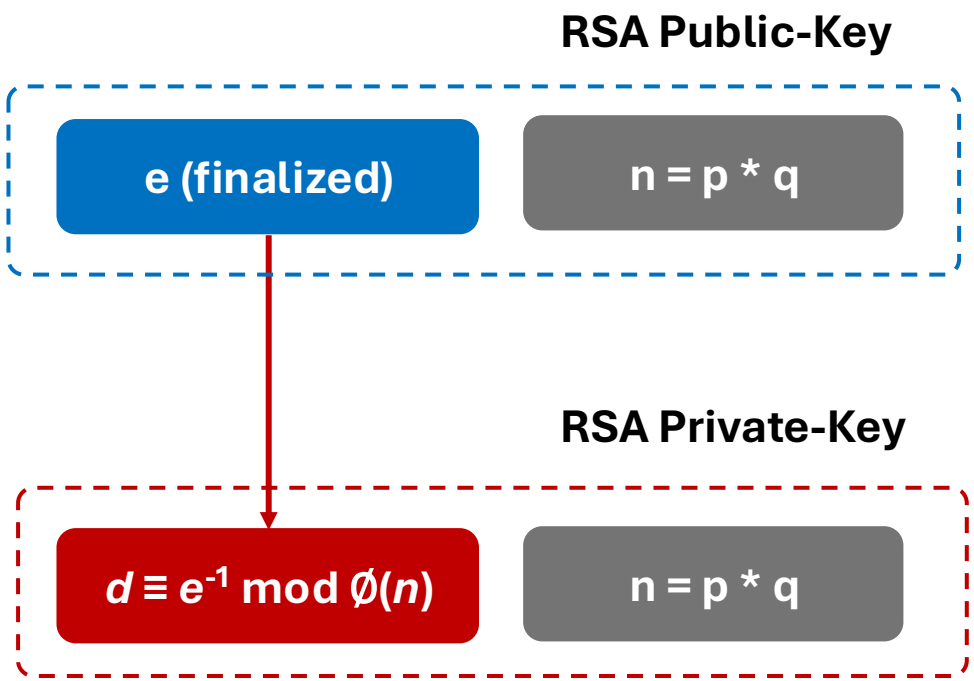
$PR = \{d, n\}$

$M = C^d \bmod n$



Summary of RSA Algorithm





Security of RSA

- Computational difficulty of factoring large prime **n**
  - If an attacker can factor n into p and q, they can calculate ϕ(n) and then d
- Size of the key: RSA security increases with key size
  - Minimum recommended **2048-bit key**
- Attacks on RSA:
  - **Brute-force Attack:** practically impossible due to the key size
  - **Factoring Attacks:** attempt to find p and q.
  - **Timing Attacks:** analyze the time taken for decryption operations.
  - **Chosen Ciphertext Attacks (CCA):** Exploit properties of RSA if not properly padded (e.g., using OAEP).

Security of RSA: Factoring Problem

- **RSA-250 (829 bits)** is the largest RSA challenge number factored so far (2020)

RSA-250 Status: Factored (Feb 2020)

829 bits / 250 digit (decimal):

2140324650240744961264423072839333563008
6147151447550177977549208814180234471
4013664334551909580467961099285187247091
4587687396261921557363047454770520805
1190564931066876915900197594056934574522
3058932597669747168173806936489469
9871578494975937497937

Approximately **2700 CPU core-years**, using a 2.1 GHz Intel Xeon Gold 6130 CPU

Factors:

6413528947707158027879019017057738908482501
4742943447208116859632024532344630238623598752668347708737661925585694639798853367

and

3337202759497815655622601060535511422794076034476755466678452098702384172921003708025744867329688187756571898625803693206271 1

Other Asymmetric Cryptosystems

- **Elliptic Curve Cryptography (ECC)**
  - Same security level as RSA, but requires smaller key size making it more efficient
  - Instead of prime factorization problem, security of ECC relies on solving discrete logarithm problem
  - Widely used in modern protocols, e.g., TLS1.3, Elliptic Curve Diffie-Hellman (ECDH) key exchange, ECDSA for digital signature
- **Elgamal Encryption**
  - Cryptographic algorithm based on the Diffie–Hellman key exchange
  - Slower than RSA for encryption and faster for decryption
  - Less common, used in PGP, theoretical interest

Topics

- Public-Key Encryption (PKE)
- Fundamental Mathematic
- RSA Algorithm
- **Key Exchange Algorithm**

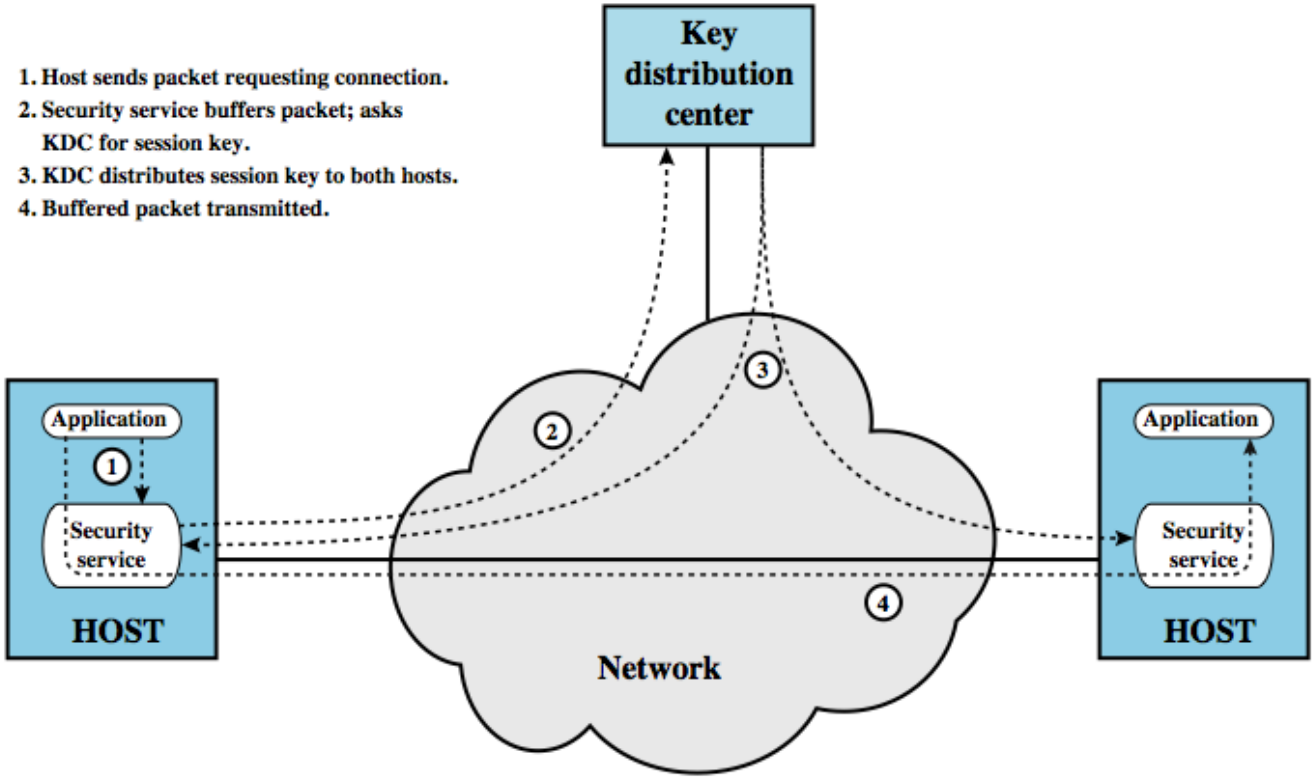
Applications of PKE

- PKE is <u>much slower</u> than **symmetric-key encryption**
- Applications of PKE in practice
  1) **Key Exchange: key transport vs key agreement**
  2) Message Authentication/Message Integrity
  3) Digital Signatures
  4) Digital Certificates

How to securely <u>exchange/establish the secret key</u> between two parties ?

Traditional Key Exchange: KDC

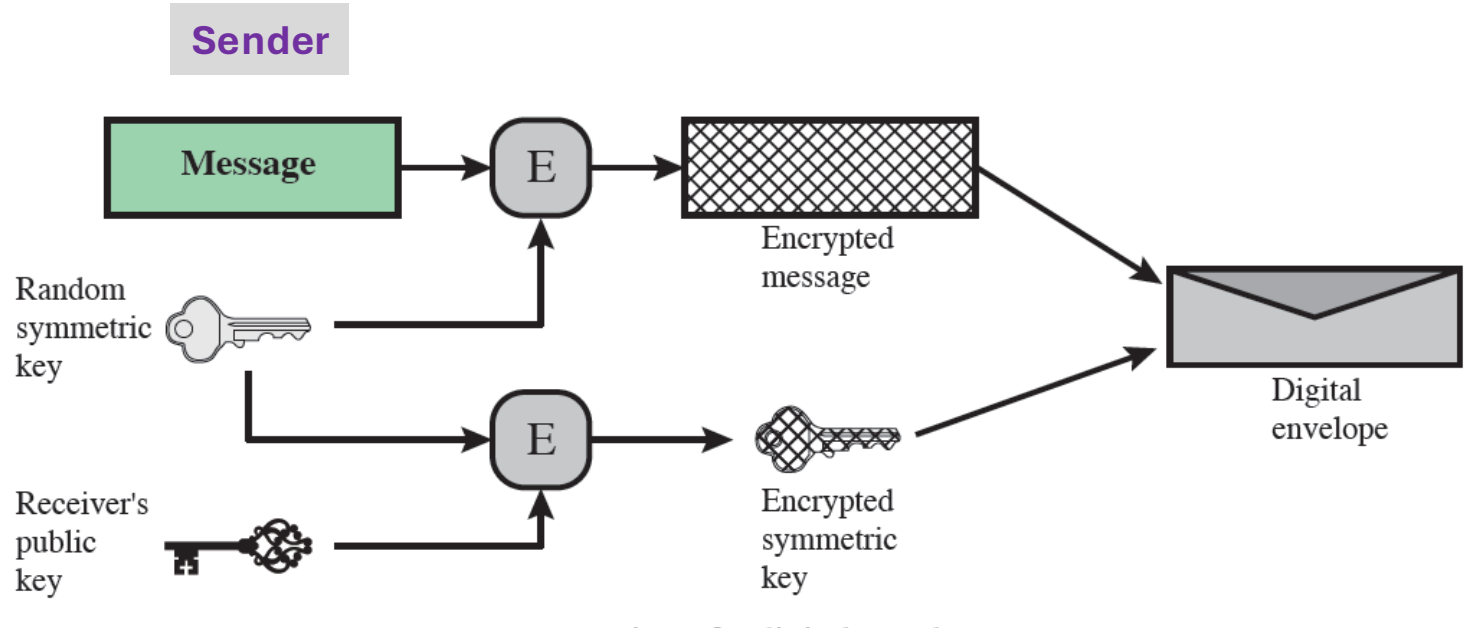
- Traditional way of exchanging key
- Using ***Key Distribution Center (KDC)***
- Rely on <u>trusted third party</u>
- Distributed session key is encrypted with each host pre-shared secret key
- **<u>Limitations:</u>** require every party to have pre-shared secret key with the central KDC



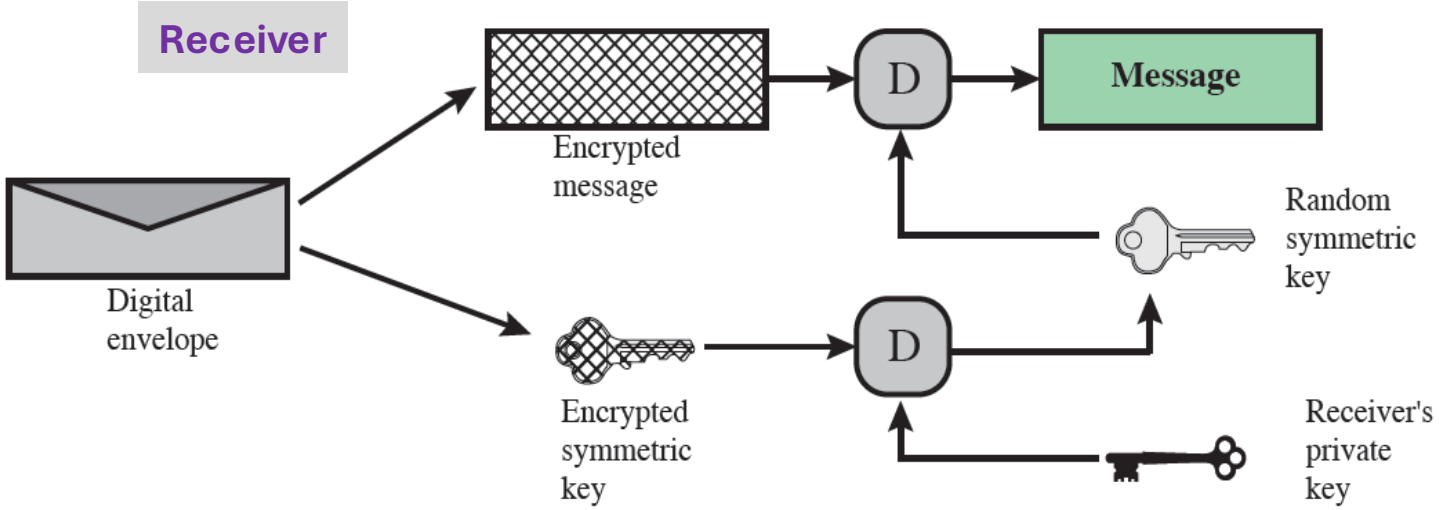
Key Exchange: Key Transport

PKE for Key Transport:

- Sender securely share **symmetric key** (for data encryption) with receiver
- Sender uses **receiver's public key** to <u>encrypt</u> **symmetric key**
- Receiver uses its own **private key** to <u>decrypt</u> **symmetric key**



Creation of Digital Envelope



Opening of Digital Envelope

Key Exchnage: Key Transport Protocol

Key Transport: one party **generates the key** and **securely sends** it to the other

Encryption (by sender) :

1. Choose random x
2. Encrypt $y = \mathrm{RSA}(e, x) = x^e \bmod n$
   (Asymmetric-key Encryption)
3. Compute session key: $k = H(x)$
   (H(?): Hash function)
4. $c = \mathrm{E}(k, m)$
   (Symmetric-key Encryption)
5. Output ciphertext (y, c)

Decryption (by receiver) :

1. $x = \mathrm{RSA}(d, y) = y^d \bmod n$
   (Asymmetric-key Decryption)
2. Compute session key : $k' = \mathrm{H}(x')$
3. $m = \mathrm{D}(k, c)$
   (Symmetric-key Decryption)

Note: Can replace RSA with ECC for key transport

Key Exchange: D-H Algorithm

- Known as **"Diffie-Hellman Key Exchange Algorithm"**
- Proposed by **Diffie and Hellman in 1976** along with the exposition of PKE concept
  - Cannot be used to exchange arbitrary message
  - On the other hand, it can **help establish a common key**
- D-H is a ***key agreement*** protocol for public exchange of a secret key
  - Both parties **contribute information** to derive the **same shared key**, without explicitly sending the key itself.
- Used in several commercial products
  - Usually integrated into larger protocols, e.g., TLS/SSL, SSH, IPSec

Primitive Root

- Let **p** be a **prime number**.
- An integer **g** is called a **primitive root modulo p** if: $g^1, g^2, g^3, \ldots, g^{\{p-1\}}$ ***(mod p)***
- produce <u>all non-zero integers</u> **modulo p** exactly once: **{1, 2, 3, .. p-1}**
- {1, 2, 3, .. p-1} are <u>coprime</u> to **p**
- ***Is 2 a primitive root of mod 7?***

<u>Example</u>: **p = 7,**

then the number **coprime to 7** is **{1,2,3,4,5,6}**

Let's test **g = 3**

- 3^1 mod 7 = 3
- 3^2 mod 7 = 2
- 3^3 mod 7 = 6
- 3^4 mod 7 = 4
- 3^5 mod 7 = 5
- 3^6 mod 7 = 1

The powers of 3 generate all numbers from 1 to 6 → so **3 is a primitive root mod 7**

How D-H Algorithm Works (1)

- Public Parameters (agreed upon by both parties)
  - **p**: a large prime number
  - **g:** a primitive root modulo p
- Each party generates their own secret key
  - Chooses a secret integer (private): **x,** where **x < p**
  - Compute a public key : $y = g^x \bmod p$
- Two parties: A and B
  - A: Keep secret integer x_A and publish public key y_A
  - B: Keep secret integer x_B and publish public key y_B

How D-H Algorithm Works (2)

- A <u>shared session key</u> for A and B is denoted as K_{AB}

$$
\begin{aligned}
K_{AB} &= Y_A^{X_B} \bmod p \\
&= Y_B^{X_A} \bmod p \\
&= g^{X_A X_B} \bmod p
\end{aligned}
$$

- K_{AB} is used as a session key in symmetric-key encryption scheme between A and B
- If A and B subsequently communicate, they will have the <u>same key as before</u> (unless they choose new public-keys)

How D-H Algorithm Works?

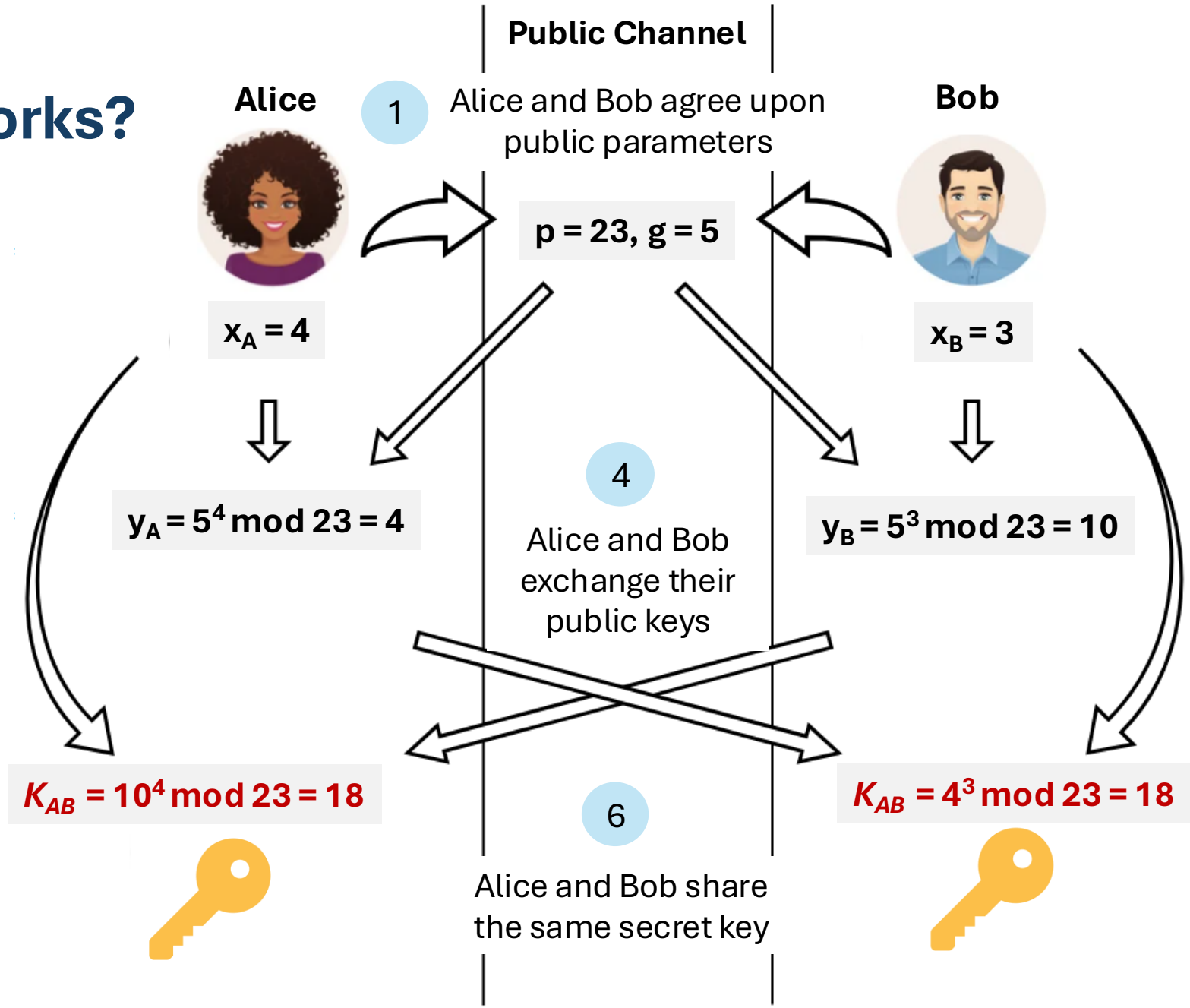


Image: https://en.wikipedia.org/wiki/Diffie%E2%80%93Hellman_key_exchange

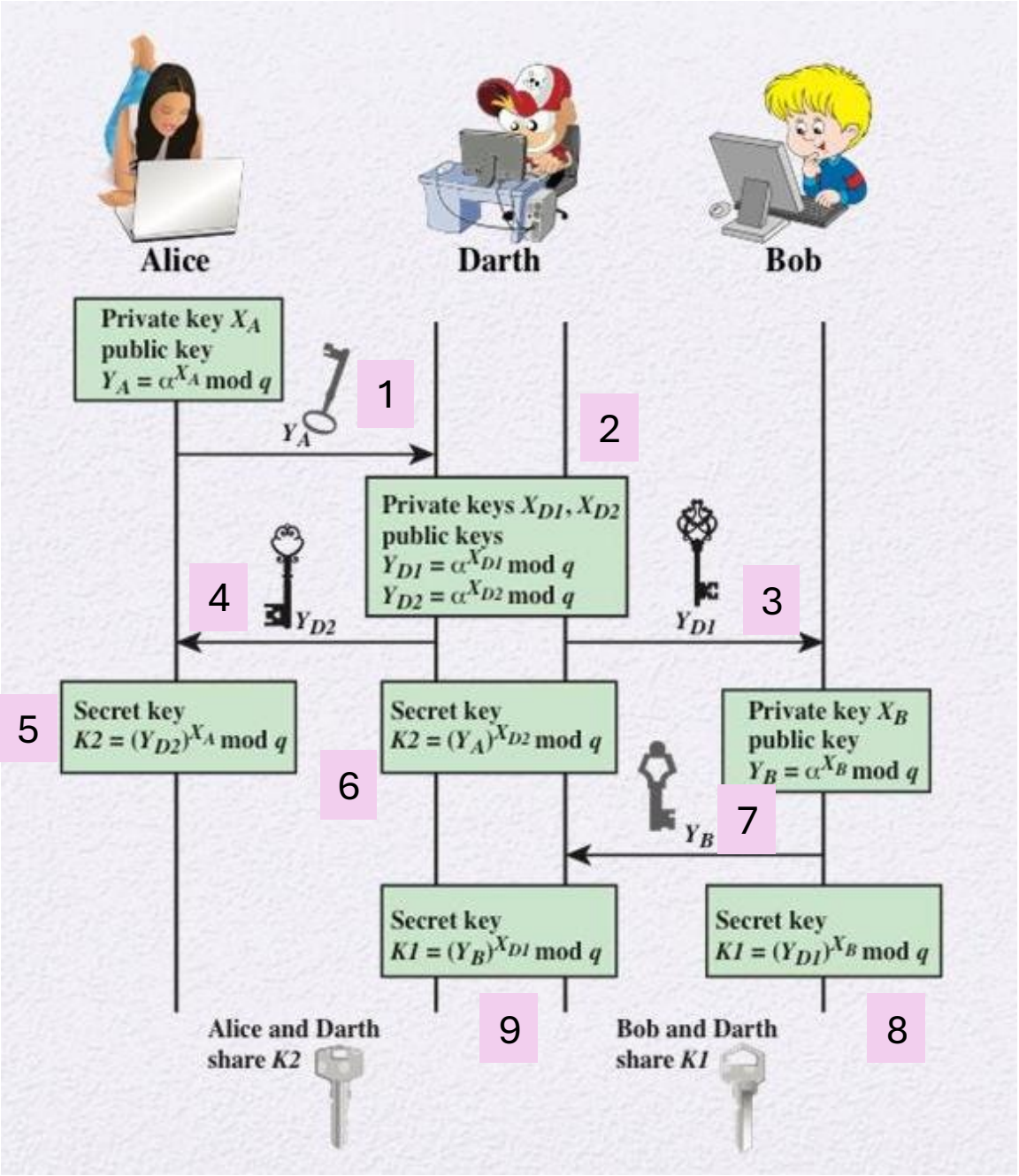
Security of D-H Algorithm

- An eavesdropper, Eve, knows the public parameters **p** and **g**, as well as the public values y_A and y_B.
  - To find the shared secret, Eve would need to find either Alice's private key x_A from $y_A = g^{X_A} \bmod p$ <u>or</u> Bob's private key x_B from $y_B = g^{X_B} \bmod p$
  - This is known as the **discrete logarithm problem**; for a large prime p, it is computationally infeasible to solve
- Diffie-Hellman is considered secure against <u>passive</u> eavesdropping
- Security depends on <u>proper parameters</u>, i.e., **p** should be large (at least 2048 bits), **g** must be primitive root, and **x** must be truly random and relatively large
- But, vulnerable to a ***Man-in-the-Middle (MITM)*** attack → **no authentication**

Security of D-H Algorithm: MITM

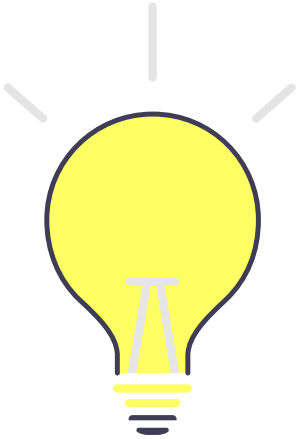
Darth can intercept all messages between Alice and Bob.

- Alice and Bob computes and sends their public key $\mathbf{Y_A}$ and $\mathbf{Y_B}$
- Darth intercepts $\mathbf{Y_A}$ and $\mathbf{Y_B}$
- Darth creates two public keys and sends one to Bob $\mathbf{Y_{D1}}$ and one to Alice $\mathbf{Y_{D1}}$
- Two different shared keys are created
  - **K1**:between Bob and Darth
  - **K2**: between Alice and Darth



Summary

- **Publick key encryption (PKE)** or **Asymmetric key encryption**
  - Require a pair of **private** and **public** keys
  - Depend on hard math problem → difficult to break
  - Slower and more complex than symmetric key encryption
- Based on number theory: prime and co-prime and modular arithmetic
- **RSA** algorithm is an example of PKE
- PKE is used for <u>key exchange</u>
  - **Key transport**: use RSA, ECC, etc., to transmit shared secret key
  - **Key agreement: Diffie-Hellman algorithm**



มหาวิทยาลัยมหิดล
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ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วนโดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

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